

Part Two: Factor Analysis and Structural Equation Modeling

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1 What Part Two Is

Part Two of *Quantitative Methods for Social and Behavioral Research with R* is about latent-variable modeling. The methods of Part One worked with variables that were recorded: a test score, a count, a group label. Here the variable of primary interest is one that no column of the data set holds. Twelve questionnaire items are recorded; the planning, the learning strategies and the sense of belonging those items are meant to register are not. Part Two asks what follows from taking that situation seriously, and answers through factor analysis: how observed responses represent latent variables, and how a measurement model is developed, specified, estimated, evaluated and extended.

It is written for postgraduate students and applied researchers who have read Part One or who already command the statistics it develops: variable form, variance, covariance and

correlation; estimates, sampling uncertainty, intervals and tests, with the limits of a dichotomous decision; regression equations, residuals and conditional comparisons; nested model comparison and penalized in-sample criteria; held-out evidence; and reproducible work in R. Part One's argument-based reading of a study is useful background rather than a prerequisite. One optional section of Chapter 1 applies it to the latent-variable case, and nothing later depends on it.

Four chapters form the measurement sequence, and this introduction is about them. Chapter 1, *From Latent-Variable Modeling to Factor Analysis*, is the general and conceptual presentation. Chapter 2, *Exploratory Factor Analysis and Model Development*, is the exploratory approach, conceptual and practical. Chapter 3, *Specifying and Estimating Confirmatory Factor Models*, is the foundational confirmatory approach, covering concepts, specification and estimation. Chapter 4, *Evaluating and Comparing Confirmatory Factor Models*, is the advanced confirmatory approach, covering evaluation, comparison and the higher-order and bifactor extensions. The chapters after them carry the structural half of the subject, from relations between factors and indicators to relations among variables in a structural system; the last section here says what they cover and what they cannot confer.

Part Two asks for more mathematics than Part One did. Readers manipulate equations, count the moments a covariance matrix supplies against the parameters a model spends, and work with small vectors and matrices: a matrix product, a transpose, a symmetric matrix, a diagonal matrix. No previous course in factor algebra or matrix calculus is assumed. Each new operation appears first in a scalar or two-variable case, substantive and graphical meaning comes before notation, and every symbol, index and dimension is defined before it is used. Calculus and proof-based mathematical statistics are not prerequisites, and long proofs, general identification theorems, extended matrix derivations, asymptotic distribution theory and hand optimization are placed so that a reader can pass over them without losing a definition, an equation or an assumption the main route needs.

The book chapters carry the statistics. A model appears as equations, as parameter matrices, as a path diagram and as an explanation; every reported number is computed for the reader and read in prose and in named tables, and no chapter displays R code. The software that specifies, fits and extracts each model belongs to the Computer Lab beside the chapter, where every model a chapter reports has a task that fits it. A reader who never opens R can still follow every model, every count and every conclusion from the printed evidence. Readers who meet these chapters in a course may also receive companion materials – lecture notes, exercise handouts, homework, and Computer Labs with worked answers – but the chapters stand on their own.

Each chapter opens in the same way. A **Chapter overview** box summarizes the whole chapter in ordinary words: the route, and what a reader will be able to do at the end. The prose that follows it begins from a real measurement problem or concern in an applied study and states the question the chapter answers. Each chapter closes with a numbered **Summary and reading bridge** that gathers what has been settled and states the questions the next chapter owes. A short **Key terms** table follows where a chapter carries one, then answer-free exercises,

then any appendices, which hold derivations, data documentation and extensions the main route does not require.

2 The Measurement Problem at a Glance

Several recorded responses vary together, and the attribute used to explain their association is not recorded at all. The common-factor model proposes an account of that covariation: each recorded response is written as a common part, shared with the other responses through a small number of unobserved variables, plus a residual part that the declared common variables do not represent. That sentence is the model. The algebra makes it precise and gives it consequences that data can contradict.

For p indicators and m factors, with $\mathbf{y}_i$ the $p \times 1$ vector of responses recorded for unit i ,

$$\begin{aligned} \mathbf{y}_i &= \nu + \Lambda \eta_i + \epsilon_i, & \mu(\vartheta) &= \nu + \Lambda \alpha, \\ & & \Sigma(\vartheta) &= \Lambda \Phi \Lambda^\top + \Theta, \end{aligned}$$

with $E(\epsilon_i) = \mathbf{0}$ and $\text{Cov}(\eta_i, \epsilon_i) = \mathbf{0}$. Here ν holds the p indicator intercepts, η_i the m factors for unit i , Λ the $p \times m$ loadings that relate the two, and ϵ_i the p residuals; α holds the m latent intercepts, Φ is the $m \times m$ factor covariance matrix, Θ the $p \times p$ residual covariance matrix, and ϑ the complete vector of free parameters. What the model produces is the implied pair: a mean structure $\mu(\vartheta)$ and a covariance structure $\Sigma(\vartheta)$. An analysis working from covariances alone sets the mean structure aside or profiles it out under stated assumptions, which is a choice rather than an absence. One notation serves all four chapters, and it is chosen to survive the move to structural models, so that later chapters extend the model rather than quietly changing what its symbols mean.

The covariance equation is what makes the proposal answerable. Declared loadings, factor relations and residual restrictions imply one particular covariance matrix, and that matrix can be held against the one a sample supplies. The comparison has force because a covariance matrix for p indicators holds only $p(p+1)/2$ distinct entries: a model that spends fewer parameters than that makes claims which can fail. Counting the information indicators supply against the parameters a model spends is a recurring operation in Part Two, and it is kept separate from the further question of whether those moments determine the parameters at all.

None of this makes a factor identical to the construct that names it. A **construct** is a substantive idea belonging to the research question, and its meaning comes from theory, item content, response processes, the population studied and the use intended for the results, none of which a covariance matrix supplies. A **factor** is a random variable inside a statistical model, and an observed or estimated **score** is neither; the three are never treated as synonyms. The arrow a path diagram draws from a factor to an indicator states the direction of representation

the model adopts, writing each response as a function of the factors plus a residual. It is not a finding that the factor produces the response, and it is the wrong direction for a construct whose indicators produce it. Fit to covariance evidence becomes construct and score language only when item content, response processes, population and intended use also support that reading.

Part Two models continuous indicators and continuous outcomes. Ordered categorical indicators, thresholds, polychoric correlations, ordered-response factor models and their estimator-specific conventions are signposted where a reader would meet them and are taught elsewhere; numeric category codes are not automatically continuous measurements. The boundary concerns the indicators rather than every variable in a study. Categorical grouping variables and dummy-coded predictors stay in scope, and the later multigroup and MIMIC work uses them alongside continuous indicators. Full-information maximum likelihood for missing continuous responses and robust maximum likelihood for inference under nonnormality are later topics as well, and neither is a switch to categorical measurement.

The model family is the reflective common-factor measurement model with its higher-order and bifactor relatives, estimated by normal-theory maximum likelihood as one coherent lineage across exploratory and confirmatory work; principal components analysis answers a different question and is not a common-factor estimator. What the family cannot do is as settled as what it can. No factor analysis establishes that a latent entity exists, that a reflective direction is causal, that a scale is valid, or that a structural claim is warranted. Higher-order and bifactor models are rival representations of an adequate core model rather than repairs for a weak one, and reliability follows a defined score, model, population and use: it is not validity, and it does not mend a weak measurement representation.

3 How Chapters 1–4 Progress

The four chapters make four connected moves on one covariance equation: **represent** a common-factor model and derive the moments it implies; **explore** a development record to settle dimensionality, orientation and interpretation; **specify and estimate** the resulting proposal as explicit restrictions on evidence that played no part in producing it; and **evaluate, compare and extend** the fitted model. The equation is the same throughout. What changes at each stage is the responsibility it carries and the evidence it must answer to. Figure 1 sets out that architecture; no chapter redraws it.

Table 1 states what each chapter is responsible for and what it hands to the chapter after it.

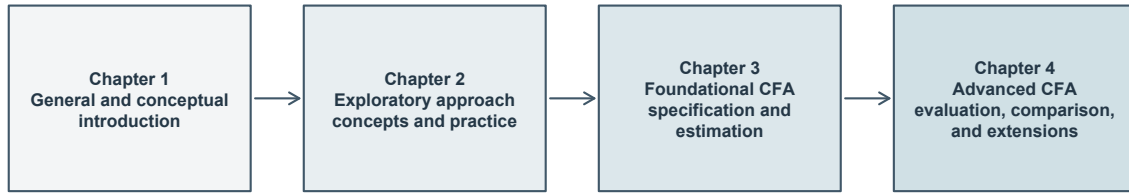


Figure 1: The four-chapter architecture moves from representation to exploratory development, foundational CFA, and advanced CFA.

Table 1

The four-chapter measurement sequence.

Chapter	Main responsibility	What it hands forward
1. From Latent-Variable Modeling to Factor Analysis	Separate an observed variable, a latent random variable, a fixed parameter and a residual, and a factor from the construct that names it; write one two-factor, six-indicator system as scalar equations, matrices, a path diagram and a sentence and check the four against one another; say what a scaling, sign, location or coordinate convention supplies and what it cannot decide; derive $\Sigma = \Lambda\Phi\Lambda^T + \Theta$; and say what exact reproduction does not establish	The vocabulary the three later chapters use without redefinition, and the known-value six-indicator records on which the algebra is checked before any sample is read

Chapter	Main responsibility	What it hands forward
2. Exploratory Factor Analysis and Model Development	Audit indicators, units, coding and missingness before reading a correlation matrix; state which matrix an extraction targets; compare observed eigenvalues with a matched permutation reference and read a structured retention profile across seven rules; count effective parameters and candidate degrees of freedom; transform loadings and factor covariance together and say what rotation preserves; and read pattern coefficients, structure correlations, factor correlations and communalities together, locating discrepancy in a residual matrix rather than an average	A frozen candidate specification with its named rivals, its retained cross-loadings, the retention evidence behind it and an explicit list of unresolved alternatives, settled before any confirmation evidence is inspected
3. Specifying and Estimating Confirmatory Factor Models	State one restricted model as words, equations, parameter matrices and a diagram and check that they agree; separate free, fixed, equality-constrained, bounded and defined entries; set a latent metric by marker or fixed-factor scaling and transform between them; trace a cross-loading and a residual covariance into the moments each changes; count moments, free parameters and degrees of freedom and say why the count is not identification; and check convergence, admissibility, gradients and local rank before reading any coefficient	An evaluation record built in memory and never written to disk: the exact specification and parameter ledger, the active items in order, the moment and degrees-of-freedom audit, the estimator and scaling settings, the convergence and admissibility record, the estimates with their uncertainty, and the rivals declared in advance

Chapter	Main responsibility	What it hands forward
4. Evaluating and Comparing Confirmatory Factor Models	Reconstruct the discrepancy the estimator minimized under a named divisor convention; read the exact-fit test, RMSEA with its interval, CFI, TLI and SRMR as answers to different questions; separate the three residual scales; establish exact nesting, run a compatible difference test and say when AIC and BIC are eligible; specify, count, fit and compare the higher-order and bifactor extensions; read modification indices and expected parameter change as leads that change analytic status; and freeze one revision and evaluate it on untouched records	A calibrated measurement conclusion whose analytic history and unresolved strain stay intact, and the structural question the later chapters take up: which latent variables are endogenous, which directional restrictions enter, and how measurement uncertainty is carried forward

The stopping points are as deliberate as the moves. Chapter 2 does not relabel a developed pattern as confirmation. Chapter 3 does not treat convergence or admissibility as adequate fit, and says so in its own section title: it reads an estimated model *before its fit*. Chapter 4 does not treat favorable fit as construct validation. Each chapter completes one responsibility and transfers a defined record, rather than carrying an unexamined conclusion forward under a new name.

4 Evidence Roles Across the Sequence

Using one model family repeatedly does not give every record the same inferential role. Part Two keeps the analysis history visible so that development, first evaluation and independent revalidation are not collapsed into one favorable account. Every decision is developmental, planned in advance, a planned sensitivity, or post hoc, always in relation to a named body of evidence, and that status follows the timing and provenance of the decision rather than the name of the function that fitted it: a model developed after inspecting a record is exploratory with respect to that record whatever routine estimates it. Table 2 gives the principal roles.

Table 2

Records keep distinct analytical roles even when they illustrate the same model family.

Record	First permitted use	What it cannot establish by itself
Known-value six-indicator records	Chapter 1 reconstructs an implied covariance matrix entrywise from parameters declared in advance	Sampling behavior, an empirical factor count, a unique rotation, a best model, or the existence of the constructs the columns are named after
SLE Sample A, 360 cases, development	Chapter 2 audits the indicators, develops the exploratory solution, and freezes the candidate with its named rivals	Independent confirmation of any choice made while Sample A was being inspected
SLE Sample B, 360 cases, confirmation	Chapter 3 estimates the frozen candidate; Chapter 4 evaluates it, diagnoses it, and compares the rivals declared in advance	Independent support for a relation first noticed in Sample B, which stays exploratory until fresh evidence tests it
SLE Sample C, 360 cases, revalidation	Chapter 4 evaluates the one revision frozen after Sample B	General validity, measurement invariance, or permission to search Sample C repeatedly for a revision that survives
Known SLE generating moments	Chapter 4 discloses them only after the Sample B and Sample C sequence is complete, then computes the population optimum and the population discrepancies directly	Contemporaneous justification for an earlier choice, or a finite-sample conclusion from one realization beyond what the audit separates
Separate donor microcases	Compact demonstrations of a specified calculation or model comparison, declared from their own population and seed	The development or confirmation status of the SLE sequence, which they never join

The Student Learning Experiences (SLE) records are a synthetic course case: continuous simulated data, not observations from actual students and not a validated instrument. Three independent samples of 360 complete cases carry twelve indicators – four Planning, four Strategy and four Belonging – drawn from one declared population and distributed together in `sle_continuous.csv` beside the chapters. Each analysis uses only the sample its role entitles it to, and Chapter 1 uses none of them; the Chapter 2 appendix supplies the codebook, the sample manifest and the generating population, so that every reported value can be reproduced. The samples are never pooled to manufacture a larger analysis, availability is not permission to use a record early, and a record that becomes exploratory through revision does not recover confirmatory status because the next command is named after confirmation.

The retrospective audit that closes Chapter 4 follows the same discipline in reverse. It is possible only because the population is known, and it is disclosed only after the Sample B and Sample C sequence has run its course: first the evidence a researcher could actually have had, then the generating structure the candidate omitted. The audit adds no further sample, no new candidate and no successful revision, and it supplies no after-the-fact justification for a choice made earlier. Its value is that a reader watches a model stay coherent, reproducible and apparently well fitting while still owing a narrower conclusion than its name suggests – which no real study can show.

5 Boundaries and What Follows

The chapters after Chapter 4 continue Part Two from relations between factors and indicators to relations among variables in a structural system. Their subject is observed-variable path analysis; general structural equation modeling; measurement invariance and multigroup measurement; MIMIC models; latent regression and indirect effects; latent growth in a structural equation form; and structural model comparison, ordered by their research questions rather than by a software menu. Multilevel modeling, including multilevel growth, belongs to Part Three – a division of the curriculum, not a claim that structural and multilevel parameterizations stop being equivalent when observations are unbalanced or incomplete. Item-response theory, mixture models, multilevel factor analysis and a full treatment of categorical indicators lie outside Part Two altogether.

The measurement sequence ends where a reader can read and evaluate a measurement representation, and structural work begins where a new question requires relations among variables or among factors. An acceptable measurement model is necessary evidence for many latent-variable structural analyses, but a measurement portion plus a structural portion confers no substantive warrant: structural equation modeling does not upgrade an associational design to a causal one, does not erase measurement uncertainty, and does not make an indirect effect causal by calling it mediation. Before latent means or structural parameters are compared across groups, settings or occasions, the relevant level of measurement comparability is addressed with the loading, intercept and other restrictions it requires; and a growth model addresses both mean trajectories and the variation and covariation in trajectories. The bridge out of Chapter 4 is written with that in view, so that a latent intercept is not mistaken for the marginal latent mean once structural regressions enter.

Part One's argument vocabulary is reused rather than redefined, and Part Two does not re-teach the argument-based approach as a recurring apparatus. One optional section of Chapter 1 shows how evidence, justification, interpretation and claim boundaries change when a statistical variable is latent, applies the six quality-evaluation questions under Part One's names, records at least one failed verdict and repairs the claim. A reader may skip it without losing a definition, an equation, an exercise or a prerequisite for the chapters that follow, and Chapters 2 to 4

express prespecification, analytic status, uncertainty, limitations and claim boundaries in the ordinary technical vocabulary of exploratory and confirmatory factor analysis.

The software is allocated as deliberately as the evidence. Chapter 1's chapter text needs no student software: its records are constructed, its results exact, and its conclusions readable without a computer. The Computer Labs install the small pinned stack once, before it is needed, and then put it to work where the statistics require it – `psych` and `GPArotation` for the exploratory work from Chapter 2, and `lavaan` for the confirmatory fits from Chapter 3. A chapter names the software it relies on and states the consequential defaults in words – scale setting, factor covariances, mean structures, missing-data handling, standard errors and the test-statistic family – while the commands themselves stay in the labs.

The reference that follows is the single vocabulary for the four chapters. Its four concept tables gather related terms into broad concept groups and give each term's meaning or essential distinction; a closing table records the notation the chapters use. In the concept tables the **Main explanation** column is a destination, the chapter and section where the term is principally explained, with a first use elsewhere noted where one exists; in the notation table the **Main use** column carries that destination and a final column records alternative symbols and the cautions attached to them. Terms that name the same concept in context share an entry, related concepts with different meanings have separate entries, and a symbol whose preferred form carries a qualifier takes a row of its own that names its other uses. First-use explanations, worked examples and the reasons behind each distinction stay in the chapters themselves; the reference is a place to return to as they develop the ideas.

Reference: Central Concepts in Chapters 1–4

Latent Variables and the Common-Factor Model

Table 3

Latent variables and the common-factor model.

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Latent-variable modeling	Latent variable (common factor, η_k)	An unobserved random variable introduced by a model and given a distribution over units. It is not a parameter, not a score, and not made observable by being named.	Chapter 1, Sec. 1.1
Latent-variable modeling	Construct	A substantive idea in the research question. Its meaning comes from theory, item content, response processes, population and intended use, none of which a covariance matrix supplies.	Chapter 1, Sec. 1.1

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Latent-variable modeling	Indicator (item, observed response Y_j)	An observed variable proposed to bear on a factor; an item response is the value recorded for one unit.	Chapter 1, Sec. 1.1
Latent-variable modeling	Domain sampling (content or behavioral domain coverage)	Defensible coverage of the domain a construct names. It supports a broad construct interpretation and is usually systematic rather than a literal probability sample; it is a different question from sampling units from a population.	Chapter 1, Sec. 1.1
Latent-variable modeling	Reflective direction (formative alternative)	The modelling choice that writes each indicator as a function of the factors plus a residual. It states a direction of representation, not a causal finding, and it fails for constructs whose indicators produce them.	Chapter 1, Sec. 1.1
Latent-variable modeling	Classical true score ($X_{ir} = T_i + E_{ir}$)	The expected score of a person over repetitions of a specified procedure. It is a property of a person and a procedure, so a stable irrelevant influence stays inside it; it equals a common factor only under added assumptions.	Chapter 1, Sec. 1.2
Common-factor model	Common-factor model (linear factor model)	$\mathbf{y}_i = \nu + \Lambda\eta_i + \epsilon_i$ with the moment assumptions on η_i and ϵ_i . Its predictors are not columns of the dataset, which is what separates it from a regression.	Chapter 1, Sec. 2.1–2.2
Common-factor model	Loading (pattern coefficient, λ_{jk})	A fixed unknown parameter relating indicator j to factor k on a chosen factor metric. It is not a person's score and not a direct measure of how much attribute a person has.	Chapter 1, Sec. 2.1
Common-factor model	Residual (unique part, ϵ_j) and residual covariance θ_{jk}	The part of an indicator the declared factors do not represent, which may hold random fluctuation, item-specific content, method or occasion effects, an omitted factor, or misspecification. A residual covariance is a pair-specific relation outside the common part.	Chapter 1, Sec. 2.3
Common-factor model	Factor covariance matrix (Φ)	The $m \times m$ variance-covariance matrix of the factors, symmetric and positive semidefinite. It is total latent covariance, distinct from a disturbance covariance in a later structural model.	Chapter 1, Sec. 2.2

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Common-factor model	Local independence (conditional independence)	Indicators being independent given the factors. It is stronger than zero conditional residual covariance, which is stronger than the zero unconditional covariance a diagonal Θ supplies; none of the three says the observed items are unrelated.	Chapter 1, Sec. 2.3
Specification	Simple structure and cross-loading	A loading pattern in which each indicator relates to one factor, the remaining entries being fixed at zero; a cross-loading is a nonzero entry outside that pattern. Each zero is a restriction, not a drawing convenience. As a rotation criterion the same pattern favours relatively few substantial relations per item and identifiable indicator groups; it is not a criterion of fit to the observed covariance matrix, and rotation does not improve F_{ML} .	Chapter 1, Sec. 3.1; as a rotation criterion Chapter 2, Sec. 5.2
Specification	Fixed, free and constrained parameter (parameter restriction)	Three distinct statuses an entry of Λ , Φ or Θ can hold: fixed at a declared value, freely estimated, or tied to another parameter. A path diagram does not distinguish them.	Chapter 1, Sec. 3.2
Factor scale	Scale setting (marker variable, fixed factor)	Supplying a metric to a factor that has none, either by fixing one loading per factor or by fixing the factor variance. Neither convention is superior, and the implied covariance among indicators is the same under both.	Chapter 1, Sec. 4.1
Factor scale	Sign indeterminacy	Reversing a factor's direction and every one of its loadings leaves every product, and therefore the implied covariance, unchanged. A negative sign is a convention, and a sign flip between equivalent solutions is not a change in results.	Chapter 1, Sec. 4.2
Factor scale	Location convention (latent intercept α)	Fixing the origin of the factors, usually by setting the single-group factor means at zero, because observed means cannot determine unrestricted intercepts and unrestricted factor means together. α is a latent intercept and coincides with $E(\eta_i)$ only without structural paths.	Chapter 1, Sec. 4.2

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Factor scale	Observed-unit change	Re-expressing recorded indicators in different units, which changes loadings, residual variances and implied covariances coherently and predictably. Nothing is indeterminate; this is not a latent-coordinate convention.	Chapter 1, Sec. 4.3
Factor scale	Rotational freedom (coordinate change, $\mathbf{Q}$)	The freedom, with two or more factors, to replace the axes while corresponding changes to Λ and Φ leave the observed covariance unchanged. It resolves a representation choice and establishes nothing substantive.	Chapter 1, Sec. 4.1 and 7
Covariance structure	Model-implied covariance $\Sigma(\vartheta)$ (covariance structure analysis)	$\Lambda\Phi\Lambda^\top + \Theta$: the covariance the model implies, split into a common and a residual part. It needs finite second moments and the stated moment restrictions, not normality.	Chapter 1, Sec. 5
Covariance structure	Pattern matrix and structure matrix ($\Lambda\Phi$)	Λ holds the coefficients of the indicator equations; $\Lambda\Phi$ holds indicator-factor covariances. They coincide under orthogonal unit-variance factors and must be read side by side in an oblique solution.	Chapter 1, Sec. 5; oblique reading Chapter 2, Sec. 6.1
Covariance structure	Communality (h_j^2)	The common part of an indicator's variance, $\lambda_j^\top \Phi \lambda_j$, under the declared model. The shortcut λ_j^2 holds only for one nonzero loading on a unit-variance factor; correlated factors add a cross term.	Chapter 1, Sec. 5.1
Covariance structure	Uniqueness (residual variance; Heywood case)	$\theta_{jj} = 1 - h_j^2$ for a standardized indicator: variance the declared common factors do not represent, which is not the same as random error. A negative estimate is an improper, or Heywood, solution.	Chapter 1, Sec. 5.1
Covariance structure	Moment structure ($\mu(\vartheta)$)	The modeled object is the mean and covariance pair; $\mu = \nu + \Lambda\alpha$. Analysing covariances alone sets the mean structure aside, which is a choice rather than an absence.	Chapter 1, Sec. 5.2
Covariance structure	Distinct moment and degrees of freedom ($u = p(p+1)/2$, $df = u - t$)	The number of distinct entries a covariance matrix supplies, and its difference from the free-parameter count. More respondents estimate the same moments more precisely; $df \geq 0$ is necessary for, and never proof of, identification.	Chapter 1, Sec. 5.3

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Covariance structure	Tetrad restriction	The vanishing of $\sigma_{jk}\sigma_{lm} - \sigma_{jm}\sigma_{lk}$, which one common factor with uncorrelated residuals forces for every four indicators. A nonzero tetrad refutes exact reproduction by that model, whatever its loadings.	Chapter 1, Sec. 6.3
Covariance structure	SS loading (column sum of squared loadings)	A summary of one factor column across indicators, read after rotation as a column statistic of the oriented solution. It is not in general an eigenvalue of the analysed correlation matrix, the matrix being decomposed must always be named, and with correlated factors it gives no unique additive partition of common variance.	Chapter 1, Sec. 6.1; post-rotation reading Chapter 2, Sec. 3.2
Neighboring objects	Factor-score estimate ($\hat{\eta}_{ik}$)	A prediction of an unobserved factor value from responses, a fitted model and a scoring rule. Different rules give different numbers from the same fit, and no rule makes it an error-free measurement.	Chapter 1, Sec. 6.5
Neighboring objects	Principal component and prespecified composite	Weighted composites of observed variables, one chosen to capture observed variance and one declared as a rule such as a sum or mean. Neither decomposes covariance into common and residual parts, so neither answers the common-factor model's question.	Chapter 1, Sec. 6.5
Analytic orientation	Exploratory, confirmatory and hybrid orientation	Names for how much is declared before the focal analysis, not for unrelated models. Both use the same covariance decomposition; the restrictions and the history of the decisions separate them.	Chapter 1, Sec. 1.4 and 6.6
Analytic orientation	Analytic status (development versus confirmation evidence)	The record of which body of evidence informed each decision. A specification chosen on one body of evidence receives a genuinely new test only from evidence not used to choose it; analyses move from confirmatory to exploratory status and never back.	Chapter 1, Sec. 6.6–6.7

Exploratory Factor Analysis and Model Development

Table 4

Exploratory factor analysis and model development.

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Exploratory analysis	Exploratory factor analysis (EFA)	A common-factor analysis that leaves most loading restrictions open while seeking a comparatively low-dimensional representation of shared covariation. It is not assumption-free: the indicator set, population, association matrix, residual restriction, dimensional range and rotation criterion all shape what it can reveal.	Chapter 2, Sec. 1
Exploratory analysis	Development sample (development record)	The sample whose inspection produces a model. A model developed after inspecting these observations is exploratory with respect to them whatever fitting function is used, so the same records cannot confirm it.	Chapter 2, Sec. 1
Extraction	Squared multiple correlation (SMC)	For indicator j , the R^2 from regressing it on all other indicators, computed as $1 - 1/(\mathbf{R}^{-1})_{jj}$. It is a regression summary of the observed matrix used as a communality start, not a known common variance.	Chapter 2, Sec. 2.3
Extraction	Reduced correlation matrix (SMC-reduced matrix)	The correlation matrix with communality estimates replacing the unit diagonal. It need not be positive semidefinite, and its eigenvalues are on a different scale from the full matrix's.	Chapter 2, Sec. 2.3
Extraction	Principal-axis factoring (PAF)	Iterative extraction that decomposes a reduced matrix, forms loadings from its leading positive roots and updates the diagonal from row sums of squared loadings. The diagonal start, nonpositive-root handling and stopping rule are implementation choices.	Chapter 2, Sec. 2.3
Extraction	Minimum-residual extraction (minres)	Extraction minimizing the sum of squared off-diagonal correlation residuals, giving each displayed discrepancy equal weight. It is a common-factor method but not another name for PAF.	Chapter 2, Sec. 2.4

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Retention	Kaiser–Guttman rule (eigenvalue-greater-than-one)	Retaining roots above one in a unit-diagonal correlation analysis. Sampling variation alone produces roots above one, and applying the threshold to a reduced matrix changes the comparison.	Chapter 2, Sec. 3.3
Retention	Scree plot	An ordered plot of eigenvalues read for where large decreases give way to a flatter sequence. It supplies no sampling reference, and full- and reduced-matrix plots need not agree.	Chapter 2, Sec. 3.3
Retention	Parallel analysis (permutation parallel analysis)	Comparing each observed eigenvalue with the same-rank eigenvalues of datasets built to have no common factors but the same sample size and marginal distributions, and retaining the leading consecutive run that clears its reference. The eigenvalue basis, the null generator, the reference statistic and the replication count and seed all belong to the specification.	Chapter 2, Sec. 3.4
Retention	Empirical Kaiser criterion (EKC)	A sample-adjusted repair of Kaiser–Guttman using the Marchenko–Pastur bound $(1 + \sqrt{p/n})^2$ for the first root, correcting later roots for variance already accounted for and flooring the reference at one.	Chapter 2, Sec. 3.5
Retention	Sequential model tests	Fitting successive factor counts and stopping at the first non-significant exact-fit test. The sequence is trustworthy only while still under-fitting, because the χ^2 reference fails once the count exceeds the true number.	Chapter 2, Sec. 3.5
Retention	Combination rule	Using two criteria with different references together, and treating their disagreement as evidence that the dimensions are hard to detect rather than as a tie to be broken by vote.	Chapter 2, Sec. 3.5
Counting	Effective EFA parameter count (t_m)	$pm + p - m(m - 1)/2$ for an unrestricted common-factor covariance model at regular points, with $df_m = p(p + 1)/2 - t_m$. The subtraction removes rotational redundancy; an oblique rotation adds no parameters because it re-expresses the same fitted covariance.	Chapter 2, Sec. 4.1

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Counting	Ledermann bound	The largest factor count the covariance moments can generically support, $m \leq (2p + 1 - \sqrt{8p + 1})/2$. It is a counting limit, not a recommendation, a positive-root count or a proof of identification.	Chapter 2, Sec. 4.1
Rotation	Inverse-paired transformation	The general nonsingular change of factor coordinates $\Lambda \mathbf{T}^{-1}$ with $\mathbf{T} \Phi \mathbf{T}^T$, which leaves the fitted common covariance unchanged. Multiplying loadings while leaving Φ alone is not an equivalent oblique rotation.	Chapter 2, Sec. 5.1
Rotation	Varimax	An orthogonal criterion maximizing the dispersion of squared loadings within columns. A clean varimax pattern does not establish that the substantive attributes are unrelated.	Chapter 2, Sec. 5.2
Rotation	Direct oblimin (quartimin at $\gamma = 0$)	An oblique criterion discouraging large squared coefficients in several columns of one row while permitting factor correlations. The name alone is not a specification: γ , normalization, tolerance, iteration limit, starts and seed all matter.	Chapter 2, Sec. 5.2
Rotation	Kaiser normalization	Dividing each loading row by its communality-based length before optimizing and restoring the scale afterward, which reweights indicators in the criterion. It is not the same operation as standardizing the observed variables.	Chapter 2, Sec. 5.2
Reading an oblique solution	Pattern coefficient (pattern matrix $\mathbf{P}$)	The coefficient of a factor in the simultaneous decomposition of a standardized indicator, read conditional on the other factors in the same representation.	Chapter 2, Sec. 6.1
Reading an oblique solution	Structure correlation (structure matrix $\mathbf{C} = \mathbf{P} \Phi$)	The zero-order indicator–factor correlation, which can be appreciable where the pattern coefficient is near zero because part of the relation travels through the factor correlation. Pattern and structure coincide only in an orthogonal solution.	Chapter 2, Sec. 6.1 (defined as $\Lambda \Phi$ in Chapter 1, Sec. 5)
Reading an oblique solution	Oblique communality as a quadratic form	$h_j^2 = \mathbf{p}_j^T \Phi \mathbf{p}_j$, whose factor-correlation cross-products are part of the common variance. A row sum of squared pattern coefficients omits them and is a different quantity.	Chapter 2, Sec. 6.2

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Residual evidence	Correlation residual matrix (E) and off-diagonal RMS	Observed minus reproduced correlations, summarized by $\sqrt{2 \sum_{j < k} e_{jk}^2 / [p(p-1)]}$, which excludes variances. The largest absolute residual and the location of discrepancy carry information an average conceals, and this summary is not automatically any software statistic named RMSR or SRMR.	Chapter 2, Sec. 7.1
Residual evidence	Uniqueness boundary (bound-constrained uniqueness)	A fitted uniqueness at the implementation's lower bound rather than in the interior. It can signal too many factors, a redundant pair, sampling fluctuation or misspecification, and is different evidence from an interior estimate.	Chapter 2, Sec. 4.3
Preliminary summaries	KMO (Kaiser–Meyer–Olkin measure of sampling adequacy)	The ratio of summed squared zero-order correlations to that sum plus the summed squared partial correlations. It describes the input; it selects no factor count and is no minimum-sample-size calculation.	Chapter 2, Sec. 7.2
Preliminary summaries	Bartlett's test of sphericity	A normal-theory test of $H_0 : \mathbf{R}_{\text{population}} = \mathbf{I}$ on $p(p-1)/2$ df. Rejection says the indicators are not mutually uncorrelated, not that a low-dimensional factor model fits.	Chapter 2, Sec. 7.2
Analytic sensitivity	Tucker's congruence coefficient	The cosine between two loading-column vectors, compared after column order and sign are aligned. It compares coefficient directions, not participant scores, and near-one congruence can hide an important difference for one item.	Chapter 2, Sec. 7.3
Analytic sensitivity	Analytic sensitivity comparison	Changing one declared analytic choice at a time on the same records and comparing aligned solutions. It is same-sample evidence about analytic robustness, never an independent replication or a sampling interval.	Chapter 2, Sec. 7.3
Analytic status	Frozen candidate specification	The exact item order, free and fixed relations, scaling, covariance and mean settings, estimation family and named rivals recorded before any confirmation evidence is inspected. Freezing is what makes a later comparison interpretable.	Chapter 2, Sec. 8

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Analytic status	Planned rival model	A restriction or broader alternative named before the confirmation outcome, distinguished from a post-hoc change, which changes analytic status and requires new evidence.	Chapter 2, Sec. 8

Confirmatory Specification, Identification, and Estimation

Table 5

Confirmatory specification, identification, and estimation.

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Confirmatory specification	Confirmatory factor analysis (restricted factor model; CFA)	A common-factor model whose loading, residual, and factor-covariance entries are restricted before estimation, giving testable implications beyond the number of factors. The label describes restrictions and analytic history, not a favorable conclusion or a function name.	Chapter 3, Sec. 1
Confirmatory specification	Exclusion restriction (fixed zero loading; absent arrow)	A loading entry set to zero as a substantive claim that the indicator has no direct relation to that factor. It does not state that the indicator and factor are uncorrelated, because correlated factors still induce a nonzero structure coefficient.	Chapter 3, Sec. 1
Parameter status	Free, fixed, equality-constrained, bounded, and defined parameter	Estimated under the other restrictions; assigned a constant; required to share a value; restricted to a range; or computed from fitted parameters without adding freedom. A diagram shows which relations exist, not which status each has.	Chapter 3, Sec. 3
Parameter status	Residual covariance (correlated residual; correlated uniqueness)	A freely estimated off-diagonal entry of Θ , occupying two symmetric cells and adding one free parameter. It changes one indicator pair's implied covariance, unlike a cross-loading, which changes a whole row and column.	Chapter 3, Sec. 3
Latent metric	Marker scaling (reference indicator; unit loading identification)	Fixing one loading per factor at one, so the factor is expressed in that indicator's units and the factor variance is estimated. It does not set the factor variance to one and does not make the marker error-free.	Chapter 3, Sec. 4

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Latent metric	Fixed-factor scaling (unit variance identification; <code>std.lv</code> = TRUE)	Fixing each factor variance at one and estimating every target loading, so a factor unit is one model standard deviation in a single-group model. It does not standardize the observed indicators.	Chapter 3, Sec. 4
Latent metric	Factor-scale reparameterization ($\mathbf{D}$ transformation)	The diagonal rescaling $\Lambda^* = \Lambda \mathbf{D}^{-1}$, $\Phi^* = \mathbf{D} \Phi \mathbf{D}^\top$, which leaves the implied covariance and Θ unchanged. Equivalence fails if a constraint is not transformed with the metric or a factor variance is on a boundary.	Chapter 3, Sec. 4
Observed units	Observed-unit rescaling (Δ rescaling)	Replacing $\mathbf{y}$ by $\mathbf{b} + \Delta \mathbf{y}$, which multiplies an indicator's intercept and loading by its factor, its variance and residual variance by that factor squared, and its covariance with an unchanged indicator by the factor. It changes the record, not the latent coordinates, and leaves fit unchanged under a consistently transformed model.	Chapter 3, Sec. 4
Observed units	Correlation reparameterization (standardized-indicator model)	Writing $\mathbf{R} = \mathbf{D}_y^{-1/2} \Sigma \mathbf{D}_y^{-1/2}$ with rescaled loading and residual matrices. It standardizes the indicators, not the factors, and its moment count differs from the covariance form because the unit diagonal is fixed.	Chapter 3, Sec. 4
Coefficient metrics	Latent-standardized and fully standardized loading (<code>std.lv</code> ; <code>std.all</code>)	$\lambda \sqrt{\phi_{aa}}$ expresses a factor change in standard-deviation units in the indicator's raw unit; $\lambda \sqrt{\phi_{aa}} / \sqrt{\sigma_{jj}}$ standardizes both. A raw standard error is not the standardized coefficient's standard error, because the transformation uses estimated quantities.	Chapter 3, Sec. 4
Moment accounting	Observed moments, free parameters, and model degrees of freedom (u , t , $df = u - t$)	The distinct sample moments a declared representation supplies, the free coordinates the model estimates, and their difference. Necessary bookkeeping for one stated representation; neither identification nor quality, and unique covariance entries are not observations.	Chapter 3, Sec. 6
Moment accounting	Unrestricted-EFA dimension	The count of distinct covariance-model axes in an unrestricted m -factor solution, $pm + p - m(m - 1)/2$ at regular full-rank points, which subtracts the continuous rotational freedom that changes the display without changing the reproduced covariance.	Chapter 3, Sec. 6

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Identification	Identification (global, local, and generic; empirical or weak identification)	Whether the moments determine the parameters: uniquely over the admissible space, in a neighborhood, or except at exceptional values. Weak identification describes a recoverable model whose parameters are poorly distinguished, showing as unstable estimates and very large standard errors rather than an error message.	Chapter 3, Sec. 6
Identification	Two- and three-indicator sufficient conditions (connectedness)	A factor with at least three exclusive indicators is identified under a fixed scale and diagonal residuals; a factor with two exclusive indicators needs a nonzero covariance with another identified factor. Both are sufficient and neither is necessary, and the second depends on an estimated quantity.	Chapter 3, Sec. 6
Identification	Moment Jacobian rank check	Full column rank of $\partial \mathbf{s} / \partial \vartheta^T$ at a regular fitted point, a local diagnostic that supports local identification. It does not prove global uniqueness or uniform stability, and it is a check on the fitted solution rather than a structural proof.	Chapter 3, Sec. 6
Model relations	Nested, nonnested, and equivalent models (observational equivalence)	Nesting requires that every covariance matrix one model generates belongs to the other's set, shown by the restrictions producing the inclusion. Equal degrees of freedom imply neither nesting nor equivalence, and observational equivalence limits what covariance data can distinguish.	Chapter 3, Sec. 6
Estimation	Normal-theory ML discrepancy (F_{ML})	$\log \Sigma(\vartheta) + \text{tr}\{\mathbf{S}\Sigma(\vartheta)^{-1}\} - \log \mathbf{S} - p$, minimized over the admissible parameter space and normalized to zero at $\Sigma = \mathbf{S}$. The covariance equation needs no normality; the conventional standard errors and test do.	Chapter 3, Sec. 7 (first used Chapter 2, Sec. 2.4)
Estimation	Covariance divisor convention ($\mathbf{S}_N$ against $\mathbf{S}_{N-1}$; normal against Wishart)	$\mathbf{S}_N = \{(N-1)/N\}\mathbf{S}_{N-1}$ is the normal-likelihood convention; the Wishart convention uses an $N-1$ covariance and multiplier. A supplied covariance matrix communicates neither the sample size nor the divisor, so the convention must be disclosed and never applied twice.	Chapter 3, Sec. 7

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Mean structure	Indicator intercepts and the latent location convention	Free indicator intercepts with latent intercepts fixed at zero add p moments and p parameters, leaving the covariance restrictions and degrees of freedom unchanged. Freeing both at once identifies no latent origin, because $\alpha + \mathbf{c}$ with $\nu - \Lambda \mathbf{c}$ reproduces the same means.	Chapter 3, Sec. 5
Solution quality	Convergence, admissibility, and the Heywood case	Convergence is an optimizer's stopping statement; admissibility requires positive residual variances and positive definite factor and implied covariance matrices. A negative or zero residual variance is a symptom of the whole decomposition and is neither a cosmetic warning nor something to replace by zero in a report.	Chapter 3, Sec. 8
Inference	Wald test of a linear restriction ($\mathbf{R}\boldsymbol{\vartheta} = \mathbf{r}$)	A χ^2_q test computed from the estimates and their covariance matrix, requiring the covariance between estimates and not only their variances. It tests the stated restriction conditional on the rest of the model; it is not a global-fit test and need not equal a refitted likelihood-ratio statistic.	Chapter 3, Sec. 8
Transition record	Evaluation transition record (handoff record)	What evaluation begins from: evidence identity, specification, parameter conventions, estimation settings, data handling, numerical status, parameter output, fitted moments, stored evaluation fields, and reproduction metadata. Storing those fields is not interpreting them.	Chapter 3, Sec. 8
Observed system	Parceling (item aggregation)	Fitting $\mathbf{z} = \mathbf{A}\mathbf{y}$ with $\text{Var}(\mathbf{z}) = \mathbf{A}\Sigma\mathbf{A}^\top$. Reducing dimension lets different item-level patterns give the same parcel covariance, so a well-fitting parcel model does not establish the item-level restrictions.	Chapter 3, Appendix

Evaluation, Comparison, Reliability, and Extensions

Table 6

Evaluation, comparison, reliability, and extensions.

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Global fit	Model chi-square (exact-fit test; likelihood-ratio model test)	$T_M = N\widehat{F}_{ML}$ referred to $\chi^2_{df_M}$, testing whether some admissible parameter vector reproduces the population moment structure exactly. Nonrejection is not acceptance of the model, and rejection at large N is evidence against exact fit rather than a measure of its magnitude.	Chapter 4, Sec. 2
Global fit	RMSEA (root mean square error of approximation) with its 90% interval	Estimated discrepancy per degree of freedom, $\sqrt{\max\{(T_M - df_M)/(N df_M), 0\}}$, comparable across models of different size, with an interval obtained by inverting a noncentral chi-square reference. A zero estimate is a truncation, not zero residuals; the convention (N or $N - 1$) must be named.	Chapter 4, Sec. 2
Global fit	Incremental fit index (CFI; TLI or NNFI) with its independence baseline	Improvement of the model relative to a stated reference rather than relative to exact fit. The usual baseline frees the observed variances and fixes all covariances to zero, so a high value depends on how poor that baseline is; TLI is not bounded by one.	Chapter 4, Sec. 2
Global fit	SRMR (standardized root mean square residual)	Root mean square of observed-SD-standardized covariance residuals over one triangle including the diagonal. Not a mean absolute residual; a small value can hide a few large cells among many near-zero ones.	Chapter 4, Sec. 3
Local diagnosis	Covariance residual (raw; observed-SD-standardized; correlation residual)	Observed minus implied covariance, either in the product of the two indicators' units, divided by the product of observed SDs, or as a difference of separately formed correlations. None is a residual divided by its sampling standard error, so none is automatically a z statistic.	Chapter 4, Sec. 3

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Local diagnosis	Modification index (score statistic; Lagrange multiplier) and expected parameter change (EPC; standardized EPC)	A local one-parameter approximation to the improvement from releasing one fixed restriction, with the approximate value the released parameter would take. Read with the residual pattern, the parameter's meaning, and the status of the search; not the exact LR change, and not an instruction.	Chapter 4, Sec. 6
Comparison	Nested difference test (χ^2 difference; likelihood-ratio difference test)	$\Delta T = T_R - T_U$ on $\Delta df = df_R - df_U$ for regularly nested, compatible ML fits on identical cases, outcomes, scale, mean structure, and likelihood convention. A smaller parameter count is not proof of nesting, and failure to reject is not equivalence.	Chapter 4, Sec. 4
Comparison	Information criteria for covariance structures (AIC; BIC) eligibility	$-2\ell + 2q$ and $-2\ell + q \log N$ with q the effective free-parameter count. They rank eligible candidates fitted to the same observations and outcomes under one likelihood convention, including nonnested ones; neither is a test or an adequacy index.	Chapter 4, Sec. 4
Comparison	Score, Wald, and likelihood-ratio views of one restriction	Three statistics for one null: slope and information at the restricted fit; distance of the unrestricted estimate from the restriction relative to its uncertainty; difference of optimized likelihoods. Asymptotic equivalence is not finite-sample equality.	Chapter 4, Sec. 4
Comparison	Equivalent models (covariance-equivalent representation)	Distinct specifications that imply the same fitted covariance matrix, and hence the same statistic, df, and every fit index. No amount of covariance data can choose between them; only theory and evidence beyond reproduced covariance can.	Chapter 4, Sec. 5
Extensions	Higher-order factor model (second-order factor; first-order disturbance)	A model in which one further latent variable accounts for the covariances among first-order factors, $\Phi_l = \gamma\gamma^\top + \Psi_h$. The first-order disturbances are not indicator residuals, and the form imposes $m(m-3)/2$ restrictions, hence none when $m = 3$.	Chapter 4, Sec. 5

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Extensions	Bifactor model (general plus orthogonal specific or group factors)	Every item loads directly on one general factor and one assigned specific factor, all factors mutually orthogonal. A specific factor is common variance among its items after the general factor is held constant, not an item residual, and the model is generally nonnested against a correlated-factor model of the same items.	Chapter 4, Sec. 5
Extensions	Schmid-Leiman transformation and the proportionality restriction	The algebraic rewriting of a higher-order model as an orthogonal general factor with loadings $\Lambda_l\gamma$ and specific factors with loadings $\Lambda_l\Psi_h^{1/2}$. Because both come from one loading matrix, each simple-structure item in a domain splits its common variance in the ratio $\gamma_k/\psi_{h,k}^{1/2}$; a freely estimated bifactor model is not bound by that ratio.	Chapter 4, Sec. 5
Extensions	Method effect; testlet effect	Common variance among items sharing a format (reverse keying, response scale, informant) or a stimulus (passage, vignette, apparatus). Both are design-known groupings and the least controversial motivation for a specific factor, which generalizes a freed residual covariance to several items.	Chapter 4, Sec. 5
Reliability	Model-based omega for a declared score, $\omega(\mathbf{w})$ (composite reliability)	The proportion of a declared weighted composite's model-implied variance attributable to all modeled common factors. It retains cross-loadings, factor correlations, and modeled residual covariances, and is conditional on model, weights, and sample.	Chapter 4, Sec. 8
Reliability	Omega-hierarchical, $\omega_h(\mathbf{w})$	The share of a declared score's variance attributable specifically to a bifactor general factor. Its complement within total omega is modeled reliable specific-factor variance, not error, and a substantial value does not establish essential unidimensionality.	Chapter 4, Sec. 5

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Reliability	Average variance extracted (AVE) and coefficient H as bounded neighbours	AVE summarizes average communality for a simple fully-standardized single-factor item set; $H = a/(1 + a)$ with $a = \lambda^\top \Theta^{-1} \lambda$ is a maximum over optimally weighted composites. Neither is the reliability of a stated unit-weighted score, and neither applies unqualified to a cross-loading subscale.	Chapter 4, Appendix
Scores	Factor score (regression or conditional-mean score; Bartlett score) and factor-score determinacy	A model-dependent linear prediction of a latent variable from the indicators, whose weights use the whole fitted structure; determinacy is the model-based correlation between factor and prediction. A score is a prediction with nonzero conditional variance, not an observed latent value.	Chapter 4, Sec. 8
Analytic history	Pseudo-true (population-ML) parameter and population discrepancy (F_0 ; population RMSEA)	The optimum of a misspecified model fitted to the exact generating covariance, and the discrepancy remaining there. Displacement at this optimum is deterministic under two stated specifications, not sampling fluctuation and not finite-sample estimator bias.	Chapter 4, Sec. 7
Analytic history	Post-selection revision and independent revalidation	A change suggested by inspecting a body of evidence is exploratory with respect to that evidence, whatever function fits it; only records not used to suggest it can support it, and a retest that fails to support it does not prove the parameter is zero.	Chapter 4, Sec. 6

Notation for Chapters 1–4

Table 7
Notation for Chapters 1–4.

Group	Symbol (preferred form)	Meaning	Main use	Alternatives and notes
Indices and counts	$i, j, k; n, p, m$	Unit, indicator and factor indices; numbers of units, indicators and factors	Chapter 1, Sec. 2.1	$p = 6$ and $m = 2$ throughout Chapter 1's running system; N writes the analysis sample size in the likelihood conventions of Chapter 3, Sec. 7 and Chapter 4, Sec. 2; a also serves as a factor index in Chapters 2–3
Observed	$y_{ij}, \mathbf{y}_i, Y_j$	One response, the $p \times 1$ response vector for unit i , and the indicator as a random variable	Chapter 1, Sec. 2.1–2.2	Uppercase only where the distinction from an observed value matters
Model	ν, ν_j	The $p \times 1$ vector of indicator intercepts	Chapter 1, Sec. 2.1–2.2	—
Model	$\Lambda, \lambda_j, \lambda_{jk}$	The $p \times m$ loading (pattern) matrix, its j th row as a column vector, and the entry relating indicator j to factor k	Chapter 1, Sec. 2.2, 3	Rows are indicators and columns are factors; Λ_{simple} for the restricted pattern; Λ_l is the first-order loading matrix of Chapter 4, Sec. 5; Part One's notation uses λ for the Poisson rate (Part One Chapters 2 and 12) and the penalty size (Part One Chapter 13), so the loading reading is stated at first use
Model	η_i, η_{ik}	The $m \times 1$ vector of common factors for unit i and one factor value	Chapter 1, Sec. 2.1	η_i^* for a rescaled, sign-reversed, shifted or rotated version; η is the GLM linear predictor in Part One's notation (Part One Chapter 11) and η^2 an ANOVA effect size (Part One Chapter 4), so the factor reading is stated at first use

Group	Symbol (preferred form)	Meaning	Main use	Alternatives and notes
Model	$\epsilon_i, \epsilon_{ij}$	The $p \times 1$ residual vector and one residual	Chapter 1, Sec. 2.1	Never called measurement error without an argument
Model	α	The $m \times 1$ latent intercept vector; the factor mean only where no structural paths exist	Chapter 1, Sec. 2.2	With $\mathbf{B}$ present, $E(\eta_i) = (\mathbf{I} - \mathbf{B})^{-1}\alpha$; no κ convention; Part One's notation uses α for the Type I error probability (Part One Chapter 3) and the elastic-net mixing parameter (Part One Chapter 13)
Model	Φ, ϕ_{jk}	The $m \times m$ factor covariance matrix and its entries; $\Phi_{2\rho}$ for the correlated two-factor case	Chapter 1, Sec. 2.2, 6.4	Total latent covariance across the series, not a disturbance covariance Ψ ; Φ_l is the first-order factor covariance matrix of Chapter 4, Sec. 5; Part One's notation uses Φ for the standard Normal distribution function (Part One Chapter 2)
Model	Θ, θ_{jk}	The $p \times p$ residual covariance matrix and its entries	Chapter 1, Sec. 2.2–2.3	$\text{diag}(\theta_{11}, \dots, \theta_{pp})$ for the diagonal working version; the parameter space is $\mathcal{P}$, never another Θ ; Part One's notation uses $\theta, \hat{\theta}$ for a generic estimand (Part One Chapter 3) and the negative-binomial dispersion (Part One Chapter 12)
Implied moments	$\vartheta, \Sigma(\vartheta), \mu(\vartheta)$	The free-parameter vector and the implied covariance and mean it produces	Chapter 1, Sec. 5	Σ alone where the argument is clear; Chapter 3, Sec. 1 reads ϑ as the complete free-parameter vector, distinct from Θ

Group	Symbol (preferred form)	Meaning	Main use	Alternatives and notes
Implied moments	h_j^2	The communality of indicator j under the declared model	Chapter 1, Sec. 5.1	$\theta_{jj} = 1 - h_j^2$ for a standardized indicator; written as the quadratic form $\mathbf{p}_j^\top \Phi \mathbf{p}_j$ in Chapter 2, Sec. 6.2 and for a cross-loading indicator in Chapter 3, Sec. 8; u_j^2 writes the matching uniqueness in Chapter 2
Counting	u, t, df	Distinct observed moments, free parameters, and their difference	Chapter 1, Sec. 5.3	df upright; $u = 21$ for six indicators; reused in Chapter 3, Sec. 6; the subscripted u_j^2 of Chapter 2 is a uniqueness, not a moment count; an unsubscripted t here is the free-parameter count, distinct from the Part One t_{df} reference distribution (Part One Chapter 3)
Counting	$u_{\text{cov}}, u_{\text{mean+cov}}$	Moment counts without and with an unrestricted mean vector	Chapter 3, Sec. 5	Free indicator intercepts with latent intercepts fixed add equally to moments and parameters, leaving df unchanged
Counting	t_m	The effective parameter count of an unrestricted m -factor solution, $pm + p - m(m - 1)/2$	Chapter 2, Sec. 4.1	The unrestricted-EFA dimension of Chapter 3, Sec. 6 is the same count; an unsubscripted t is the free-parameter count of Chapter 1, Sec. 5.3

Group	Symbol (preferred form)	Meaning	Main use	Alternatives and notes
Coordinates	$a, \mathbf{c}, \mathbf{Q}$	A nonzero rescaling constant, an $m \times 1$ location shift, and an orthogonal coordinate matrix	Chapter 1, Sec. 4.1–4.2, 7	$\mathbf{T}$ for the general invertible transformation is Chapter 2's; $\mathbf{Q}$ is the orthogonal rotation matrix of Chapter 2, Sec. 5.2; a also indexes factors in p_{ja} , λ_{ja} and ϕ_{aa} (Chapters 2–3) and names $\lambda^T \Theta^{-1} \lambda$ in coefficient H (Chapter 4, Appendix), so the setting is stated at first use
Coordinates	$\mathbf{T}$	The general nonsingular change of factor coordinates, $\Lambda \mathbf{T}^{-1}$ paired with $\mathbf{T} \Phi \mathbf{T}^T$	Chapter 2, Sec. 5.1	Reused for the rotational family in Chapter 3, Sec. 6; the orthogonal case keeps Chapter 1's $\mathbf{Q}$
Coordinates	γ, δ	The oblimin parameter and the geomin parameter of the rotation criteria	Chapter 2, Sec. 5.2	Plain letters; the bold γ and δ of Chapter 4, Sec. 5 are second-order loadings and first-order disturbances
Classical test theory	X_{ir}, T_i, E_{ir}	Observed score on repetition r , true score, and error	Chapter 1, Sec. 1.2	$E_r(\cdot i)$ averages over repetitions with the person fixed
Scores	$\hat{\eta}_{ik}$	A factor-score estimate for unit i on factor k	Chapter 1, Sec. 6.5	The hat marks a prediction from a fitted model, never the factor; $\mathbf{A}_R$ and $\mathbf{A}_B$ of Chapter 4, Appendix supply the weights
Fixed arrays	$\mathbf{I}_p, \mathbf{1}_p, \text{diag}(\cdot)$	The $p \times p$ identity, the $p \times 1$ vector of ones, and the diagonal matrix formed from the listed entries	Chapter 1, Sec. 6.1–6.2	$\mathbf{1}_6$ writes the one-factor record's equal loadings as $.70 \mathbf{1}_6$

Group	Symbol (preferred form)	Meaning	Main use	Alternatives and notes
Sample records	$\mathbf{S}, \mathbf{R}$	The sample covariance matrix, divisor $n - 1$, and the sample Pearson correlation matrix	Chapter 2	Reserved in Chapter 1, which has no sample and uses $\mathbf{S}$ only for its Computer Lab and rescaling anchor; the structure matrix is written $\Lambda\Phi$ and given no symbol; Part One's notation homes both as observed-variable matrices
Sample records	$\mathbf{D}_S$	The diagonal matrix of sample variances	Chapter 2	$\mathbf{D}_y$ is the observed-variance counterpart in the correlation reparameterization of Chapter 3, Sec. 4
Sample records	$\mathbf{V}, \mathbf{D}, d_k$	Eigenvectors, the eigenvalue matrix, and the k th eigenvalue of $\mathbf{R}$	Chapter 2	A spectral eigenvalue matrix; the $\mathbf{D}$ of Chapter 3, Sec. 4 is the diagonal factor-scale transformation and Part One's notation uses $\mathbf{D}$ for a diagonal SD matrix, $\widehat{\mathbf{V}}$ for estimated coefficient covariance (Part One Chapter 8) and d for Cohen's d (Part One Chapter 3), so the setting is named at first use
Sample records	$\mathbf{R}_c^{(t)}$	The reduced correlation matrix at iteration t	Chapter 2, Sec. 2.3	The superscript is an iteration index, not the free-parameter count t ; the reduced matrix need not be positive semidefinite
Extraction	$\mathbf{A}, \mathbf{A}_m$	The unrotated loading matrix and its rank- m PCA counterpart	Chapter 2	$\mathbf{A}$ is the parcel weight matrix in Chapter 3, Appendix and $\mathbf{A}_R, \mathbf{A}_B$ are factor-score weight matrices in Chapter 4, Appendix; each use is declared locally

Group	Symbol (preferred form)	Meaning	Main use	Alternatives and notes
Extraction	F_{minres}	The minimum-residual extraction discrepancy	Chapter 2, Sec. 2.4	A common-factor criterion, not another name for principal-axis factoring
Retention	SS_k	The sum of squared loadings in column k	Chapter 2, Sec. 3.2	Generally not a spectral eigenvalue, and distinct from the Part One ANOVA SS family
Retention	$B, Q_{.95}$	The number of permutation replications and the 95th-percentile null reference	Chapter 2, Sec. 3.4	The bold B of Chapter 4, Sec. 8 is the latent structural path matrix
Retention	EKC_j	The empirical Kaiser reference for root j	Chapter 2, Sec. 3.5	Built on the Marchenko–Pastur bound $(1 + \sqrt{p/n})^2$ and floored at one
Retention	T_m, df_m	The corrected exploratory model statistic for m factors and its degrees of freedom	Chapter 2, Sec. 3.5	Lowercase m marks the exploratory sequence; the uppercase T_M, df_M of Chapter 4, Sec. 2 are the confirmatory model statistic and its degrees of freedom
Rotation	$\mathbf{P}, p_{ja}$	The pattern matrix and the pattern coefficient of indicator j on factor a	Chapter 2, Sec. 6.1	Λ names the same object in the model equations of Chapter 1 and Chapter 3
Rotation	$\mathbf{C}$	The structure matrix $\mathbf{C} = \mathbf{P}\Phi$ of indicator–factor correlations	Chapter 2, Sec. 6.1	Defined as $\Lambda\Phi$ in Chapter 1, Sec. 5, which gives it no symbol
Residual evidence	$\mathbf{E}, e_{jk}, \text{RMS}_{\text{off}}$	The correlation residual matrix, its entries, and the off-diagonal summary $\sqrt{2 \sum_{j < k} e_{jk}^2 / [p(p-1)]}$	Chapter 2, Sec. 7.1	In Chapter 4, Sec. 3 e_{jk} is the raw covariance residual and $e_{jk}^{(s)}, e_{jk}^{(r)}$ the standardized and correlation scales, so the scale is always named; this summary is not automatically any software RMSR or SRMR
Residual evidence	$\hat{r}_{jk}$	A reproduced correlation	Chapter 2, Sec. 7.1	The hat marks a model-implied value, not a sample statistic

Group	Symbol (preferred form)	Meaning	Main use	Alternatives and notes
Preliminary summaries	a_{jk}	A partial correlation entering the KMO ratio	Chapter 2, Sec. 7.2	Distinct from the rescaling constant a of Chapter 1, Sec. 4.1 and from the factor index a
Preliminary summaries	T_B, df_B	Bartlett's sphericity statistic and its $p(p-1)/2$ degrees of freedom	Chapter 2, Sec. 7.2	In Chapter 4, Sec. 2 the same pair names the independence-baseline statistic of the incremental indices, a different null
Analytic sensitivity	$c(\mathbf{a}, \mathbf{b})$	Tucker's congruence coefficient between two loading columns $\mathbf{a}$ and $\mathbf{b}$	Chapter 2, Sec. 7.3	Compared after column order and sign are aligned; these are loading columns, not the shift vector $\mathbf{b}$ of Chapter 3, Sec. 4
Parameter space	$\mathcal{P}$	The admissible parameter space	Chapter 3, Sec. 7	Never written Θ , which is the residual covariance matrix
Latent metric	$\mathbf{D}$ (factor scale)	The diagonal factor-scale transformation, $\Lambda^* = \Lambda \mathbf{D}^{-1}$ with $\Phi^* = \mathbf{D} \Phi \mathbf{D}^\top$	Chapter 3, Sec. 4	Carries its qualifier, because $\mathbf{D}$ is also the eigenvalue matrix of Chapter 2
Latent metric	$\lambda_{ja}^{(\text{lv})}, \lambda_{ja}^{(\text{all})}$	Latent-standardized and fully standardized loadings	Chapter 3, Sec. 4	$\lambda \sqrt{\phi_{aa}}$ and $\lambda \sqrt{\phi_{aa}} / \sqrt{\sigma_{jj}}$; a raw standard error is not the standardized coefficient's standard error
Observed units	$\Delta, \mathbf{b}$	The diagonal observed-unit conversion matrix and its shift vector, giving $\mathbf{b} + \Delta \mathbf{y}$	Chapter 3, Sec. 4	Reserved for observed-unit rescaling across Part Two and named without algebra in Chapter 1; the moment Jacobian is written $\partial \mathbf{s} / \partial \vartheta^\top$, never Δ . Chapter 3, Sec. 6.5 writes the moment Jacobian in that form, and the Chapter 4 local-rank check inherits it; Part Two keeps the two symbols separate

Group	Symbol (preferred form)	Meaning	Main use	Alternatives and notes
Observed units	$\mathbf{D}_y$	The diagonal matrix of observed variances used in the correlation reparameterization	Chapter 3, Sec. 4	$\mathbf{R} = \mathbf{D}_y^{-1/2} \Sigma \mathbf{D}_y^{-1/2}$; $\mathbf{D}_S$ of Chapter 2 is the sample counterpart
Observed units	$\mathbf{R}$ (implied correlation)	The model-implied correlation matrix	Chapter 3, Sec. 4	Carries its qualifier: an unqualified $\mathbf{R}$ is the sample correlation matrix of Chapter 2
Identification	$\mathbf{s}(\vartheta), \partial \mathbf{s} / \partial \vartheta^\top$	The stacked implied moments and their Jacobian	Chapter 3, Sec. 6	Full column rank at a regular fitted point supports local identification only
Estimation	F_{ML}	The normal-theory ML discrepancy	Chapter 3, Sec. 7 (first used Chapter 2, Sec. 2.4)	$\widehat{F}_{\text{ML}}$ at the fitted point enters $T_M = N \widehat{F}_{\text{ML}}$ in Chapter 4, Sec. 2; rotation does not change it
Estimation	$\mathbf{S}_N, \mathbf{S}_{N-1}$	The sample covariance matrix under the two divisor conventions, $\mathbf{S}_N = \{(N-1)/N\} \mathbf{S}_{N-1}$	Chapter 3, Sec. 7	$\mathbf{S}_N$ is reused in Chapter 4, Sec. 2; the convention is disclosed and never applied twice
Inference	$\mathbf{R}$ (restriction), $\mathbf{r}, q, W$	The restriction rows, their values, their number, and the Wald statistic for $\mathbf{R}\vartheta = \mathbf{r}$	Chapter 3, Sec. 8	Carries its qualifier; this q counts restrictions, while the q of Chapter 4, Sec. 4 is an effective free-parameter count
Parcels	$\mathbf{A}$ (parcel), $\mathbf{z}$	The parcel weight matrix and the parcel score vector, with $\text{Var}(\mathbf{z}) = \mathbf{A} \Sigma \mathbf{A}^\top$	Chapter 3, Appendix	Carries its qualifier beside the extraction and factor-score uses of $\mathbf{A}$
Global fit	T_M, df_M	The model test statistic $N \widehat{F}_{\text{ML}}$ and the model degrees of freedom of the declared covariance model	Chapter 4, Sec. 2	Uppercase M ; the lowercase T_m, df_m of Chapter 2, Sec. 3.5 belong to the exploratory sequence
Global fit	T_B (baseline), df_B	The independence-baseline statistic and its degrees of freedom in CFI and TLI	Chapter 4, Sec. 2	Carries its qualifier; the same pair is Bartlett's sphericity test in Chapter 2, Sec. 7.2

Group	Symbol (preferred form)	Meaning	Main use	Alternatives and notes
Local diagnosis	$e_{jk}, e_{jk}^{(s)}, e_{jk}^{(r)}$	Raw, observed-SD-standardized, and correlation covariance residuals	Chapter 4, Sec. 3	None is divided by a sampling standard error, so none is a z statistic; e_{jk} is the correlation residual in Chapter 2, Sec. 7.1, so the scale is named
Comparison	$\Delta T, \Delta df$	The nested difference statistic and its degrees of freedom	Chapter 4, Sec. 4	For regularly nested, compatible ML fits on identical cases, outcomes, scale, mean structure, and likelihood convention
Comparison	$\ell(\hat{\vartheta}), q$	The maximized log likelihood and the effective free-parameter count in $-2\ell + 2q$ and $-2\ell + q \log N$	Chapter 4, Sec. 4	This q is a parameter count; the q of Chapter 3, Sec. 8 counts restriction rows
Extensions	γ	The $m \times 1$ second-order loading vector	Chapter 4, Sec. 5	Bold; the plain γ of Chapter 2, Sec. 5.2 is the oblimin parameter
Extensions	$\delta, \Psi_h, \psi_{h,k}$	First-order disturbances, their diagonal covariance matrix, and its k th diagonal entry, with $\Phi_l = \gamma\gamma^\top + \Psi_h$	Chapter 4, Sec. 5	Bold; the plain δ of Chapter 2, Sec. 5.2 is the geomin parameter. First-order disturbances are not indicator residuals
Extensions	Λ_l, Φ_l	The first-order loading matrix and first-order factor covariance matrix of the higher-order form	Chapter 4, Sec. 5	The subscript l marks the first-order level
Extensions	$\lambda_{jg}, \lambda_{js}; G, S_{is}$	Bifactor general and specific loadings for item j ; the general factor and the specific factor for unit i in domain s	Chapter 4, Sec. 5	S_{is} is a specific factor, not an entry of the sample covariance matrix S
Reliability and scores	$\mathbf{w}, c_i$	Declared score weights and the composite for unit i	Chapter 4, Sec. 8	The weights are declared before any coefficient is calculated
Reliability and scores	$\omega(\mathbf{w}), \omega_h(\mathbf{w})$	The model-based common-variance proportion for those weights and omega-hierarchical	Chapter 4, Secs. 8 and 5	Both are conditional on model, weights, and sample
Reliability and scores	H	Coefficient H under a restrictive one-factor model, $H = a/(1 + a)$ with $a = \lambda^\top \Theta^{-1} \lambda$	Chapter 4, Appendix	Plain; Part One's notation uses the bold H for the regression hat matrix

Group	Symbol (preferred form)	Meaning	Main use	Alternatives and notes
Reliability and scores	$\mathbf{A}_R, \mathbf{A}_B$	Regression and Bartlett factor-score weight matrices	Chapter 4, Appendix	$\hat{\eta}_{ik}$ of Chapter 1, Sec. 6.5 is the resulting prediction
Population audit	Σ_0, Σ^*	The exact generating covariance and its pseudo-true fitted counterpart	Chapter 4, Sec. 7	Displacement at the pseudo-true optimum is deterministic, not sampling fluctuation
Population audit	F_0, RMSEA_0	The population discrepancy and the population RMSEA $\sqrt{F_0/df}$	Chapter 4, Sec. 7	Not the sample RMSEA formula
Structural extension	$\mathbf{B}, \Gamma, \zeta, \Psi$	Latent structural paths, exogenous links, the structural disturbance, and its covariance	Chapter 4, Sec. 8	Bold $\mathbf{B}$; the plain B of Chapter 2, Sec. 3.4 is a replication count. The bare Ψ is the structural disturbance covariance and nothing else: the Schmid-Leiman equations of Chapter 4, Sec. 5.5 write the first-order disturbance covariance as Ψ_h